Error-Based Active Disturbance Rejection Control for Parkinson's Disease Tremor Suppression in Wrist Considering Upper Limb Dynamics ⋆

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Abstract: Parkinson's disease is a neurodegenerative disorder whose tremor often persists despite pharmacological or surgical treatment. Active orthoses are a noninvasive alternative for wrist tremor suppression, but most approaches focus on single-joint compensation and neglect dynamic coupling from proximal joints. This work proposes an error-based active disturbance rejection control strategy for wrist tremor suppression using a three-degree-of-freedom upper limb model. The controller estimates and rejects total disturbances in real time, including model uncertainties and perturbations originating from proximal joints. Voluntary motion is estimated via lowpass filtering, and the method is evaluated against proportional integral derivative controllers. Simulation results over 100 trials for two different motion profiles show that the method achieves a balance between tremor reduction and control effort. Although an optimized proportional integral derivative controller achieves higher suppression under ideal conditions, it requires prior knowledge of voluntary motion and higher actuation. In contrast, the method maintains significant attenuation and robustness to parameter variations, indicating its potential for practical wearable tremor suppression systems.

Keywords: Parkinson's Disease, Tremor Suppression, Voluntary Motion Tracking, Human-Machine Systems, Active Disturbance Rejection Control, Upper Limb Modeling.

1. INTRODUCTION

Parkinson's disease (PD) is a chronic, progressive neurodegenerative disorder primarily characterized by the substantial loss of dopaminergic neurons in the substantia nigra (Meng et al., 2022; Fujikawa et al., 2023). This neurological decline results in a set of cardinal motor manifestations, including bradykinesia (slowness of movement), rigidity (stiffness), postural instability, and a characteristic unilateral rest tremor (Bötzel et al., 2014).

Pharmacological therapy (Bötzel et al., 2014; Lora-Millan et al., 2021) remains the first-line approach for tremor management, but a significant portion of patients does not respond adequately to medication (Fujikawa et al., 2023), and long-term pharmacotherapy often leads to reduced effectiveness and may induce debilitating side effects (Lora-Millan et al., 2021; Meng et al., 2022). Surgical interventions (Bötzel et al., 2014; Fujikawa et al., 2023) may be considered, but invasiveness and risks limit their applicability.

Actively controlled orthoses are an effective, non-invasive alternative (Lora-Millan et al., 2021; Meng et al., 2022), and particularly popular for wrist tremor suppression.

Several control strategies for wrist tremor suppression have been proposed in the literature. Gallego et al. (2013) design a neuroprosthesis based on Proportional Integrative (PI) control for a single degree of freedom (DoF) system of the human wrist, combining a Critically Damped g-h Filter (CDF), a Weighted Frequency Fourier Linear Combiner (WFLC), and a Kalman filter for tremor estimation. Copur et al. (2019) propose a zero-phase high-pass filter for tremor estimation alongside a repetitive control strategy for electrical stimulation-based co-contraction in a single DoF setup.

In parallel, Lekshmi and Ramachandran (2019) introduce a Proportional Integrative Derivative (PID) controller applied to a four DoF wrist model with gains optimized via a genetic algorithm. Zamanian and Richer (2019) develop an adaptive notch filter supported by an Adaptive Frequency Estimator (AFE) to track tremor frequency in the velocity signal. More recently, Qi et al. (2024) propose an Active Disturbance Rejection Control (ADRC) strategy

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using an Enhanced Bandlimited Multiple Fourier Linear Combiner (EBMFLC) for voluntary motion estimation.

While shoulder and elbow PD tremors significantly affect wrist motion, most approaches focus on single-joint models, neglecting joint coupling. To address this issue, a 3-DoF linear upper limb model (Gebai et al., 2017, 2018) is adopted in this work as a test platform.

Disturbance rejection-based control (Guo and Zhao, 2016) is expected to offer advantages due to its robustness to external perturbations and model uncertainties. This work investigates an error-based active disturbance rejection control for wrist tremor suppression under coupled dynamics. Two variations are considered, differing in how voluntary motion is estimated: one based on EBMFLC Qi et al. (2024) and another on Zero-Phase Low-Pass (ZPLP) filtering. These approaches are evaluated against two PID controllers, one tuned with Internal Model Control (IMC) and the other with offline Differential Evolution (DE) optimization plus perfect knowledge of voluntary motion. Robustness is evaluated through Monte Carlo simulations of physical parameters.

The remainder of this paper is organized as follows: Section 2 presents the upper limb dynamic model and the control framework. Section 3 describes the simulation setup. Section 4 discusses the obtained results. Section 5 concludes the paper and outlines future work.

2. THEORETICAL FRAMEWORK

2.1 Upper Limb Model

The upper limb is modeled as a planar three DoF system, where the shoulder, elbow, and wrist joints define the generalized coordinates. Following Gebai et al. (2018), muscles and tendons are represented as equivalent springs and dampers, resulting in the schematic shown in Figure 1. The system dynamics are described by

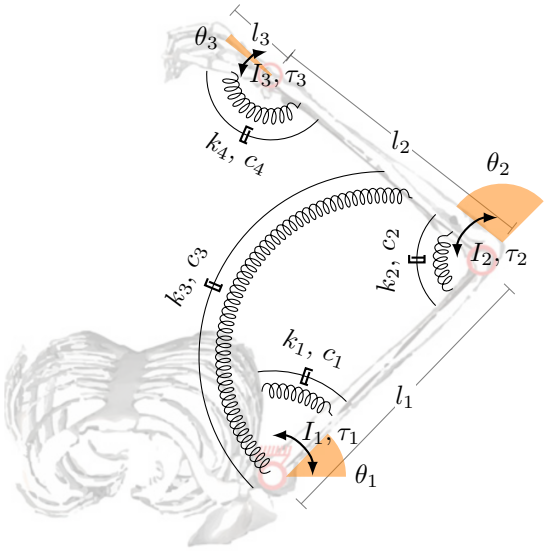
$$\mathbf{J}\ddot{\boldsymbol{\theta}} + \mathbf{C}\dot{\boldsymbol{\theta}} + \mathbf{K}\boldsymbol{\theta} = \boldsymbol{\tau}, \tag{1}$$

where $\boldsymbol{\theta} = [\theta_1 \ \theta_2 \ \theta_3]^\top$ denotes the joint flexion angles at the shoulder, elbow, and wrist, respectively, and $\boldsymbol{\tau}$ is the vector of applied torques. The linearized model is a fair representation of dynamics as long as physical parameters' change in time is small (Gebai, 2020, section 4.2.2). The inertia, stiffness, and damping matrices are given by

$$\mathbf{J} = \begin{bmatrix} j_{11} & j_{12} & j_{13} \\ & j_{22} & j_{23} \\ \text{sym.} & & j_{33} \end{bmatrix}, \quad \mathbf{K} = \begin{bmatrix} k_1+k_3 & k_3 & 0 \\ k_3 & k_2+k_3 & 0 \\ 0 & 0 & k_4 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} c_1+c_3 & c_3 & 0 \\ c_3 & c_2+c_3 & 0 \\ 0 & 0 & c_4 \end{bmatrix}. \tag{2}$$

The upper limb is composed of three rigid segments representing the arm, forearm, and hand, with masses m_1, m_2, and m_3, lengths l_1, l_2, and l_3, and moments of inertia I_1, I_2, and I_3. The distances from each joint to the segment's center of mass are denoted by $a_1 = 0.427l_1$, $a_2 = 0.417l_2$, and $a_3 = 0.361l_3$ (Gebai et al., 2018). Joint stiffness is modeled through parameters k_1, k_2, and k_4, corresponding to the shoulder, elbow, and wrist, respectively, while k_3 represents the coupling stiffness associated with the biceps. Damping coefficients c_i are assumed proportional to stiffness, such that $c_i \propto k_i$ for $i \in \{1,\ 2,\ 3,\ 4\}$.

Figure 1. Human upper limb simplified mechanical representation. Adapted from Albuquerque and Machado (2025).



The inertia matrix elements are defined as

$$j_{11} = (I_1 + m_1a_1^2) + m_2l_1^2 + m_3l_1^2 + j_{12}, \tag{3}$$
$$j_{12} = (I_2 + m_2a_2^2) + m_3(l_2^2 + l_2a_3) + j_{13}, \tag{4}$$
$$j_{13} = m_3(a_3^2 + l_2a_3), \tag{5}$$
$$j_{22} = j_{12}, \quad j_{23} = j_{13}, \quad j_{33} = I_3 + m_3a_3^2. \tag{6}$$

Finally, tremor disturbances at each joint are characterized by amplitudes A_i and frequencies f_i, for $i \in \{1, 2, 3\}$, corresponding to the shoulder, elbow, and wrist.

For a patient with PD, $\boldsymbol{\tau}$ is part voluntary, part involuntary. Denoting the pathological torque by $\boldsymbol{\tau}_i$ and the voluntary torque by $\boldsymbol{\tau}_v$, equation (7) is written (Qi et al., 2024):

$$\boldsymbol{\tau} = \boldsymbol{\tau}_i + \boldsymbol{\tau}_v. \tag{7}$$

Most of a PD tremor's power is narrowly concentrated around a central frequency (Deuschl et al., 2022). Typically, tremor is approximated by up to 3 harmonics, whose frequencies the controller attempts to identify and reject (Zamanian and Richer, 2019; Albuquerque and Machado, 2025). Since joint coupling naturally makes wrist dynamics multimodal, we've modeled $\boldsymbol{\tau}_i$ as 1 harmonic per joint:

$$\boldsymbol{\tau}_i = [A_1 \sin(2\pi f_1 t) \ \ A_2 \sin(2\pi f_2 t) \ \ A_3 \sin(2\pi f_3 t)]^\top. \tag{8}$$

By defining the state vector as $\mathbf{x} = \left[\boldsymbol{\theta}^\top \ \dot{\boldsymbol{\theta}}^\top\right]^\top$, the state space representation in (9) of the dynamic in (1) is straightforwardly obtained:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\boldsymbol{\tau} \tag{9}$$
$$\mathbf{y} = \mathbf{C}_{\text{ss}}\mathbf{x} \tag{10}$$

$$\mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{J}^{-1}\mathbf{K} & -\mathbf{J}^{-1}\mathbf{C} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} \mathbf{0} \\ \mathbf{J}^{-1} \end{bmatrix}, \quad \mathbf{C}_{\text{ss}} = [\mathbf{I}\ \mathbf{0}],$$

where $\mathbf{0}$ and $\mathbf{I}$ are 3×3 null and identity matrices.

2.2 Error-based ADRC

The ADRC is a control paradigm focused on real-time estimation and rejection of the total disturbance. This includes both internal model uncertainties and exogenous disturbances (Guo and Zhao, 2016). The classical formulation uses measurements of the plant output $y(t)$ to estimate system states and disturbances via an extended state observer. Consider a general n-th order Single-Input Single-Output (SISO) system:

$$y^{(n)}(t) = f(y, \dot{y}, \ldots, y^{(n-1)}, t) + b_0 u(t), \tag{11}$$

where $y(t)$ is the output, $u(t)$ the control input, b_0 the high-frequency gain, and $f(\cdot)$ the total disturbance.

Error-based ADRC (EADRC) defines control in terms of the tracking error $e(t) = r(t) - y(t)$ instead of the plant output, which simplifies implementation and is well suited for practical applications (Madonski et al., 2019). For the system in (11), the error dynamics are given by:

$$e^{(n)}(t) = r^{(n)}(t) - y^{(n)}(t) = r^{(n)}(t) - f(\cdot) - b_0 u(t). \tag{12}$$

Defining $f_e(t) = r^{(n)}(t) - f(\cdot)$ yields

$$e^{(n)}(t) = f_e(t) - b_0 u(t), \tag{13}$$

which retains the canonical ADRC structure in the error domain. Then, by defining the extended state vector as:

$$z_1 = e, \quad z_2 = \dot{e}, \quad \ldots, \quad z_n = e^{(n-1)}, \quad z_{n+1} = f_e, \tag{14}$$

the corresponding error-based extended state observer can be written as:

$$\begin{cases} \dot{\hat{z}}_1 = \hat{z}_2 + \beta_1 (e - \hat{z}_1), \\ \dot{\hat{z}}_2 = \hat{z}_3 + \beta_2 (e - \hat{z}_1), \\ \vdots \\ \dot{\hat{z}}_n = \hat{z}_{n+1} + \beta_n (e - \hat{z}_1) - b_0 u, \\ \dot{\hat{z}}_{n+1} = \beta_{n+1} (e - \hat{z}_1). \end{cases} \tag{15}$$

Control law is then defined as a regulator in the error coordinates:

$$u(t) = \frac{\sum_{i=1}^{n} \kappa_i \hat{z}_i(t) - \hat{z}_{n+1}(t)}{-b_0}, \tag{16}$$

where κ_i are the controller gains chosen to achieve the desired closed-loop dynamics.

Concerns While ADRC converges quickly and its implementation is simple (specially EADRC's), unstability in the observer may cause computational problems. *Ad hoc* solutions such as control signal magnitude/rate limitation are often employed in this case (Gernot Herbst, 2025).

3. METHODOLOGY

3.1 Tracking error

The observed value of wrist angle flexion θ_3 is composed by two parts, a voluntary portion due to voluntary torque from the patient, and an involuntary portion due to PD tremor, i.e.:

$$\theta_3 = \underbrace{\theta_{v_3}}_{\text{Due to } \tau_{v_3}} + \underbrace{\theta_{i_3}}_{\text{Due to } \tau_{i_3}} \tag{17}$$

since the simplified system is linear. We wish to dampen the tremor at the wrist by applying a torque u_3 to it. With respect to the modeling, this means adding a vector $\mathbf{u} = [0\ 0\ u_3]^\top$ to $\boldsymbol{\tau}$. Isolating the dynamics at the wrist (θ_3) from the state-space representation in (1), the following is derived:

$$\ddot{\theta}_3 = b_0 u_3 + \zeta(\boldsymbol{\tau}, \boldsymbol{\theta}, \dot{\boldsymbol{\theta}}), \tag{18}$$

where b_0 is the $(3,3)$ entry of $\mathbf{J}^{-1}$, and ζ is a linear combination of $\boldsymbol{\theta}$, $\dot{\boldsymbol{\theta}}$, and $\boldsymbol{\tau}$.

Ideally, the ESO tracking error would be given by

$$e_3(t) = \theta_{i_3}(t) = \theta_{v_3}(t) - \theta_3(t), \tag{19}$$

so that $e_3 \to 0$ as u_3 compensates τ_{i_3}.

In a practical setting, θ_{v_3} is unknown, and thus we estimate the tracking error using

$$\widehat{e}_3(t) = \widehat{\theta}_{v_3}(t) - \theta_3(t), \tag{20}$$

where $\widehat{\theta}_{v_3}$ is an estimate of θ_{v_3}. From (20), we have that $\widehat{e}_3 \to \epsilon_3$ as u_3 compensates τ_{i_3}, where ϵ_3 is the voluntary motion estimator error.

3.2 Simulation Setup

Controller performance is evaluated through repeated simulations under varying physical conditions. For each run, upper limb parameters are sampled around nominal values as described in Subsection 3.3, and the system response is simulated and evaluated. This procedure is repeated to assess robustness. Since involuntary torques due to Parkinson's disease are already modeled (eq. (8)), only voluntary torque profiles are specified. Voluntary torques are null to emulate resting, and to emulate motion follow the trajectories described in Section 4.

Performance is assessed using output-based and control-based metrics. For the wrist joint, Tremor Power Suppression Ratio (TPSR) (Qi et al., 2024), Integral Absolute Error (IAE) (Skogestad, 2003), Coefficient of Determination (R^2) (Glantz and Slinker, 1990), and Entropy (Lathi and Ding, 2018) are computed. Control effort is evaluated through the Total Variation (TV) (Skogestad, 2003) and Signal Power (Lathi and Ding, 2018) of the control input.

3.3 Parameters

Model Parameters Between system simulations, stiffness coefficients change according to samples drawn from uniform distributions centered at their nominal values, which also cause damping coefficients to change since they are associated to stiffness. The very first run of each control strategy is run with the nominal model, and following runs use altered models for a total of 100 runs. The 99 stiffness samples were drawn once and tabulated, so that the i-th run of every control strategy, in both scenarios, uses the same model, which avoids bias in evaluations.

Table 1 lists the nominal parameter values used in the simulations, and Table 2 defines the uniform distributions from which joint stiffness values are sampled.

Control Parameters The control parameters in Table 3 were obtained through distinct tuning methodologies. For EADRC with ZPLP, observer and controller bandwidths (w_o, w_c) were determined experimentally by maintaining the heuristic ratio $w_o = 10w_c$ ($w_o = 4w_{\text{control}}$ for EADRC

Table 1. Nominal values of simulation parameters. Based on Gebai et al. (2018).

Symbol	Value	Unit	Symbol	Value	Unit
m_1	2.07	kg	c_1	0.36	Nms/rad
m_2	1.16	kg	c_2	0.14	Nms/rad
m_3	0.54	kg	c_3	0.08	Nms/rad
I_1	0.0228	kg×m^2	c_4	0.01	Nms/rad
I_2	0.0082	kg×m^2	A_1	1.0	Nm
I_3	0.0012	kg×m^2	A_2	1.0	Nm
l_1	0.364	m	A_3	1.0	Nm
l_2	0.299	m	f_1	3.59	Hz
l_3	0.203	m	f_2	5.30	Hz
a_1	0.155	m	f_3	14.35	Hz
a_2	0.125	m	j_{11}	0.400	kg×m^2
a_3	0.073	m	j_{12}	0.102	kg×m^2
k_1	180	Nm/rad	j_{13}	0.016	kg×m^2
k_2	70	Nm/rad	j_{22}	0.102	kg×m^2
k_3	40	Nm/rad	j_{23}	0.016	kg×m^2
k_4	10	Nm/rad	j_{33}	0.004	kg×m^2

Table 2. Sampling intervals of joints stiffness.

Symbol	Min	Nominal	Max	Change	Unit
k_1	150	180	210	±16%	Nm/rad
k_2	50	70	90	±28%	Nm/rad
k_3	30	40	50	±25%	Nm/rad
k_4	5	10	15	±50%	Nm/rad

with EBMFLC). IMC-PID was tuned using the Internal Model Control method (Rivera et al., 1986), based on a second-order approximation of the wrist joint model without coupling terms. Its closed-loop time constant was set to $\lambda = f\tau$, with τ the time constant of the uncoupled wrist model and $f = 0.4$ chosen by a grid search over $f \in \{0.1, 0.2, \ldots, 5.0\}$ that maximized R^2 on the nominal model.

In contrast, DE-PID parameters were optimized via a differential evolution algorithm, following the GA-based framework in Lekshmi and Ramachandran (2019). The optimization maximized R^2 with respect to true voluntary movement. The optimization used the DE/best/1/bin strategy with a population of 15 individuals (5 per gain), a limit of 30 generations and a convergence tolerance of 10^{-3}, reached after 8 generations, and assumed full knowledge of the voluntary component θ_{v3} to establish an ideal performance benchmark. Other control approaches were adjusted according to their respective cited references.

3.4 Statistical Analysis

Each run produces matched physical-parameter samples across all four controllers, yielding a paired rather than independent design. For each performance metric, EADRC+ZPLP was compared against each controller using a paired Wilcoxon signed-rank test (Conover and Conover, 1980). A nonparametric test was chosen over the paired t-test because several metrics exhibit pronounced skew and heavy tails across runs, violating the normality assumption the t-test requires of the paired differences.

Effect sizes are reported as the rank-biserial correlation $r \in [-1, 1]$, computed from the signed ranks of the paired differences, with $|r| = 1$ indicating that one controller outperformed the other in every paired run. To control the family-wise error rate across the six metrics evaluated per scenario, p-values were adjusted using the Holm–Bonferroni procedure, and significance was assessed at 0.05.

4. RESULTS AND DISCUSSIONS

System outputs were simulated using 4th order Runge-Kutta method on the state space representation in (9). Simulation time ranged from 0s to 6s with sampling period of 1ms, with initial conditions given by (21):

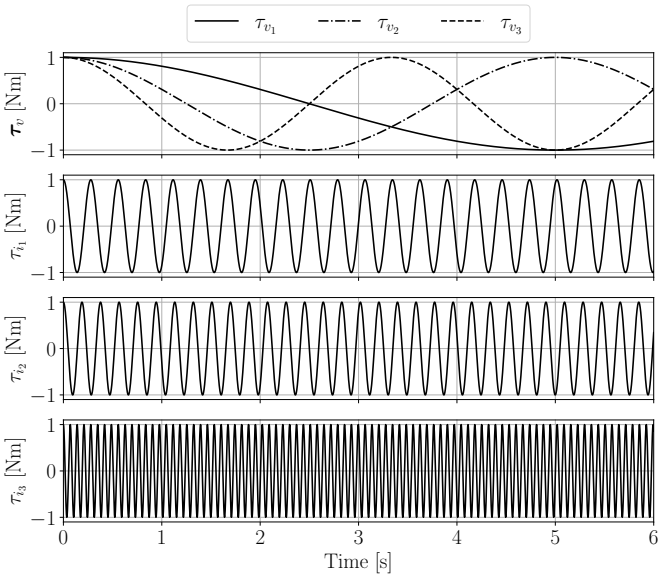
$$\boldsymbol{\theta}_0 = [0.5\ 0.1\ 0.05]^\top, \quad \dot{\boldsymbol{\theta}}_0 = [0\ 0\ 0.01]^\top. \tag{21}$$

Tremor torque $\boldsymbol{\tau}_i$ profiles, presented in Figure 2, were kept the same in all runs. The figure also presents the voluntary torques in the deliberate motion scenario, defined as:

$$\boldsymbol{\tau}_v(t) = \begin{bmatrix} \sin(2\pi \times 0.1 \times t) \\ \sin(2\pi \times 0.2 \times t) \\ \sin(2\pi \times 0.3 \times t) \end{bmatrix}. \tag{22}$$

This modeling of voluntary torque compromises between fidelity to real life and ease of implementation. Metrics from 100 simulations with varying physical parameters for $\boldsymbol{\tau}_v = \mathbf{0}$ and (22) are presented in Table 4. Figure 3 shows the effect of EADRC+ZPLP on the uncontrolled system via spectrograms.

Figure 2. Tremor torque adopted for all simulations and voluntary torque adopted for deliberate motion scenario.



Simulation results show that EADRC strategies effectively balance tremor attenuation and control effort in a coupled dynamic environment. While the DE-PID controller achieved a near-perfect TPSR above 99%, it serves mainly as an ideal benchmark, relying on full knowledge of voluntary motion and requiring significantly higher actuation. Its TV was about 3.3 times that of EADRC with ZPLP in both scenarios.

To substantiate these comparisons statistically, a paired Wilcoxon signed-rank test was applied per metric and scenario, comparing EADRC+ZPLP against each baseline controller across the same 100 Monte Carlo parameter draws.

Table 3. Controllers parameters set considering nominal model response with $\boldsymbol{\tau}$ given by (22).

Controller	Parameters				
EADRC+ZPLP	$\omega_c = 10$	$\omega_o = 100$	filter: Butterworth	cutoff: 5 Hz	order: 1
IMC-PID	$K_p = 1.235$	$K_i = 1234.660$	$K_d = 0.506$		
EADRC+EBMFLC	$n = 100$	$\mu = 0.005$	$\omega_a = 0$	$\omega_b = 8\pi$	$\omega_c = 8\pi$
	$\omega_d = 24\pi$	$T_p = 0.59$	$\alpha = 0.02$	$\omega_{\text{control}} = 20$	$\omega_o = 80$
DE-PID	$K_p = 1.721$	$K_i = 15.059$	$K_d = 3.239$		

Table 4. Average performance metrics of each control strategy.

	Controller	Response Signal				Control Signal	
		TPSR [%]	R^2	Entropy [bits]	IAE [rad×s]	Power [(N×m)^2]	TV [N×m×s]
$\boldsymbol{\tau_v = 0}$	EADRC+EBMFLC	45.92 ± 54.00	0.41 ± 0.15	5.18 ± 0.16	0.83 ± 0.28	0.50 ± 0.26	307.44 ± 89.98
	EADRC+ZPLP	57.49 ± 35.50	0.46 ± 0.20	5.17 ± 0.19	0.76 ± 0.20	0.34 ± 0.12	263.51 ± 47.82
	DE-PID	99.90 ± 0.10	1.00 ± 0.00	4.85 ± 0.10	0.03 ± 0.01	0.98 ± 0.38	877.68 ± 198.91
	IMC-PID	-123.32 ± 563.17	-0.37 ± 1.85	5.50 ± 0.31	1.24 ± 1.03	8.75 ± 25.33	492.03 ± 260.60
$\boldsymbol{\tau_v}$ as in (22)	EADRC+EBMFLC	45.02 ± 54.72	0.46 ± 0.12	5.23 ± 0.13	0.84 ± 0.28	0.51 ± 0.26	308.50 ± 89.95
	EADRC+ZPLP	56.97 ± 35.77	0.51 ± 0.16	5.21 ± 0.15	0.76 ± 0.20	0.35 ± 0.12	264.31 ± 47.75
	DE-PID	99.90 ± 0.10	1.00 ± 0.00	4.92 ± 0.09	0.03 ± 0.01	0.98 ± 0.38	877.68 ± 198.91
	IMC-PID	-154.62 ± 684.56	-0.40 ± 2.17	5.50 ± 0.29	1.31 ± 1.12	10.39 ± 31.63	511.02 ± 295.24

According to Table 5, in both scenarios EADRC+ZPLP outperforms EADRC+EBMFLC on every metric but Entropy ($p < 0.001$): it achieves higher tremor suppression (TPSR, $r \geq 0.56$; IAE, $r \leq -0.68$) while requiring less control effort (TV, $r \leq -0.81$), with lower control Power in every one of the 100 paired trials ($r = -1.00$), supporting the claimed balance between attenuation and actuation. Entropy did not differ significantly in the resting scenario ($p = 0.73$) and was moderately lower for EADRC+ZPLP in the moving one ($r = -0.32$).

Differences against DE-PID were significant for every metric in both scenarios with maximal or near-maximal effect size ($|r| \geq 0.99$), reflecting that DE-PID suppressed tremor better than EADRC+ZPLP in practically every one of the 100 paired trials, at the cost of higher control effort in all of them. This is consistent with its role as an idealized upper-bound benchmark rather than a practical competitor.

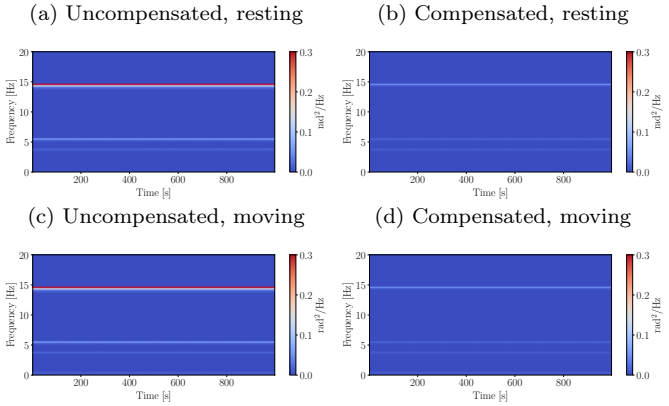
EADRC+ZPLP suppressed tremor better than IMC-PID in nearly every paired trial ($|r| \geq 0.94$ for TPSR, R^2 and IAE) while requiring less control effort in every one of them (Power and TV, $r = -1.00$). IMC-PID, tuned on the uncoupled nominal wrist, showed growing oscillations in 4 (resting) and 5 (moving) of the 99 perturbed models, which explains its negative average TPSR despite median values above 50%.

While EADRC+ZPLP has been successful in suppressing tremor regardless of the voluntary motion estimator employed, it demonstrated somewhat poor capacity in mitigating lower-frequency tremors, as spectrograms in Figure 3 show. This could indicate that the filtering process in voluntary motion estimation was not sufficient, and EADRC did not factor unmitigated tremor as part of the total disturbance.

Table 5. Holm-Bonferroni corrected p-value and rank-biserial correlation r for paired Wilcoxon signed-rank tests.

	Metric	EADRC+ZPLP vs. ⋯					
		EADRC+EBMFLC		DE-PID		IMC-PID	
		p	r	p	r	p	r
$\boldsymbol{\tau_v = 0}$	TPSR [%]	<0.001	+0.56	<0.001	-1.00	<0.001	+0.94
	R^2	<0.001	+0.60	<0.001	-1.00	<0.001	+0.94
	Entropy [bits]	0.365	-0.10	<0.001	+1.00	<0.001	-0.79
	IAE [rad×s]	<0.001	-0.68	<0.001	+1.00	<0.001	-0.88
	Power [(N×m)^2]	<0.001	-1.00	<0.001	-1.00	<0.001	-1.00
	TV [N×m×s]	<0.001	-0.81	<0.001	-1.00	<0.001	-1.00
$\boldsymbol{\tau_v}$ as in (22)	TPSR [%]	<0.001	+0.58	<0.001	-1.00	<0.001	+0.98
	R^2	<0.001	+0.65	<0.001	-1.00	<0.001	+0.98
	Entropy [bits]	0.006	-0.32	<0.001	+1.00	<0.001	-0.76
	IAE [rad×s]	<0.001	-0.69	<0.001	+1.00	<0.001	-0.95
	Power [(N×m)^2]	<0.001	-1.00	<0.001	-1.00	<0.001	-1.00
	TV [N×m×s]	<0.001	-0.82	<0.001	-1.00	<0.001	-1.00

Figure 3. Spectrograms of the uncompensated and compensated tremor of both scenarios.



5. CONCLUSION

This work investigated error-based active disturbance rejection control for wrist tremor suppression in Parkin-

son's disease while accounting for upper limb dynamics. A 3-DoF mass-spring-damper model enabled the evaluation of control performance under disturbances arising from shoulder and elbow tremors. The results show that EADRC strategies, including EBMFLC and ZPLP variations, achieve high tremor power suppression with lower control effort than the more demanding DE-PID approach. Of the two, EADRC+ZPLP achieved higher tremor suppression with smoother control signals. In particular, EADRC+ZPLP also reached an average tremor power suppression rate of 56.97% in dynamic scenarios, outperforming classical controllers.

EADRC demonstrated strong robustness with both voluntary motion estimators used, maintaining consistent performance under variations of up to 50% in joint stiffness and damping parameters. This indicates its suitability for practical applications where biomechanical properties are uncertain or time-varying. EADRC's ease of implementation and flexibility not withstanding, voluntary motion estimation is still key in avoiding the inclusion of lower-frequency tremors in the total estimated disturbance. Future work will consider the inclusion of more control strategies from the literature, as well as the investigation of alternative voluntary motion estimation methods, and use of different upper limb models.

REFERENCES

Albuquerque, G.G.d.S.B.d. and Machado, M.R. (2025). Design of passive dynamic absorbers to attenuate pathological tremor of human upper limb. In *Advances in Structural Vibration: Selected works from COBEM 2023*, 271–285. Springer Nature Switzerland, Cham. doi:10.1007/978-3-031-71540-2_20.

Bötzel, K., Tronnier, V., and Gasser, T. (2014). The differential diagnosis and treatment of tremor. *Dtsch Arztebl Int*, 111(13), 225–236. doi:10.3238/arztebl.2014.0225.

Conover, W.J. and Conover, W.J. (1980). *Practical nonparametric statistics*, volume 493. Wiley New York.

Copur, E.H., Freeman, C.T., Chu, B., and Laila, D.S. (2019). Repetitive control of electrical stimulation for tremor suppression. *IEEE Transactions on Control Systems Technology*, 27(2), 540–552. doi:10.1109/tcst.2017.2771327.

Deuschl, G., Becktepe, J.S., Dirkx, M., Haubenberger, D., Hassan, A., Helmich, R.C., Muthuraman, M., Panyakaew, P., Schwingenschuh, P., Zeuner, K.E., and Elble, R.J. (2022). The clinical and electrophysiological investigation of tremor. *Clinical Neurophysiology*, 136, 93–129. doi:10.1016/j.clinph.2022.01.004.

Fujikawa, J., Morigaki, R., Yamamoto, N., Nakanishi, H., Oda, T., Izumi, Y., and Takagi, Y. (2023). Diagnosis and treatment of tremor in parkinson's disease using mechanical devices. *Life*, 13(1). doi:10.3390/life13010078.

Gallego, J.Á., Rocon, E., Belda-Lois, J.M., and Pons, J.L. (2013). A neuroprosthesis for tremor management through the control of muscle co-contraction. *Journal of NeuroEngineering and Rehabilitation*, 10(1), 36. doi:10.1186/1743-0003-10-36.

Gebai, S., Hammoud, M., Hallal, A., and Al Shaer, A. (2017). Involuntary tremor controlled using mechanical means: Involuntary hand tremor. In *2017 Fourth International Conference on Advances in Biomedical Engineering (ICABME)*, 1–4. doi:10.1109/icabme.2017.8167537.

Gebai, S., Hammoud, M., and Khachfe, H. (2018). Parametric study of an enhanced passive absorber used for tremor suppression. *Structural Control and Health Monitoring*, 25(7), e2177. doi:10.1002/stc.2177. E2177 STC-17-0313.R2.

Gebai, S.S. (2020). *Optimization of passive cantilever-type tuned mass damper to reduce the hand postural tremor*. Ph.D. thesis, Paris Est. doi:10.70675/994e0848ze136z4e48z8db8z9981215dcb5b.

Gernot Herbst, R.M. (2025). *Active Disturbance Rejection Control: From Principles to Practice*. Birkhäuser Cham, 1 edition. doi:10.1007/978-3-031-72687-3.

Glantz, S.A. and Slinker, B.K. (1990). *Primer of Applied Regression and Analysis of Variance*. McGraw-Hill, United States of America, 1 edition.

Guo, B.Z. and Zhao, Z.L. (2016). *Active Disturbance Rejection Control for Nonlinear Systems: An Introduction*. John Wiley & Sons, Singapore, 1 edition.

Lathi, B.P. and Ding, Z. (2018). *Modern Digital and Analog Communication*. Oxford University Press, United Kingdom, 5 edition.

Lekshmi, A. and Ramachandran, K.I. (2019). Parkinson's tremor suppression using active vibration control method. *IOP Conference Series: Materials Science and Engineering*, 577(1), 012056. doi:10.1088/1757-899x/577/1/012056.

Lora-Millan, J.S., Delgado-Oleas, G., Benito-León, J., and Rocon, E. (2021). A review on wearable technologies for tremor suppression. *Frontiers in Neurology*, Volume 12 - 2021. doi:10.3389/fneur.2021.700600.

Madonski, R., Shao, S., Zhang, H., Gao, Z., Yang, J., and Li, S. (2019). General error-based active disturbance rejection control for swift industrial implementations. *Control Engineering Practice*, 84, 218–229. doi:10.1016/j.conengprac.2018.11.021.

Meng, L., Jin, M., Zhu, X., and Ming, D. (2022). Peripherical electrical stimulation for parkinsonian tremor: A systematic review. *Frontiers in Aging Neuroscience*, Volume 14 - 2022. doi:10.3389/fnagi.2022.795454.

Qi, P., Yang, J., Dai, Y., Zhang, H., and Tong, J. (2024). A novel approach to parkinson's tremor suppression: Ebmflc and ladrc integration. *IEEE Access*, 12, 145871–145880. doi:10.1109/access.2024.3424395.

Rivera, D.E., Morari, M., and Skogestad, S. (1986). Internal model control: Pid controller design. *Industrial & Engineering Chemistry Process Design and Development*, 25(1), 252–265.

Skogestad, S. (2003). Simple analytic rules for model reduction and pid controller tuning. *Journal of Process Control*, 13(4), 291–309.

Zamanian, A.H. and Richer, E. (2019). Adaptive notch filter for pathological tremor suppression using permanent magnet linear motor. *Mechatronics*, 63, 102273. doi:10.1016/j.mechatronics.2019.102273.